Signal & Image processing

BILD 62

Today's Objectives

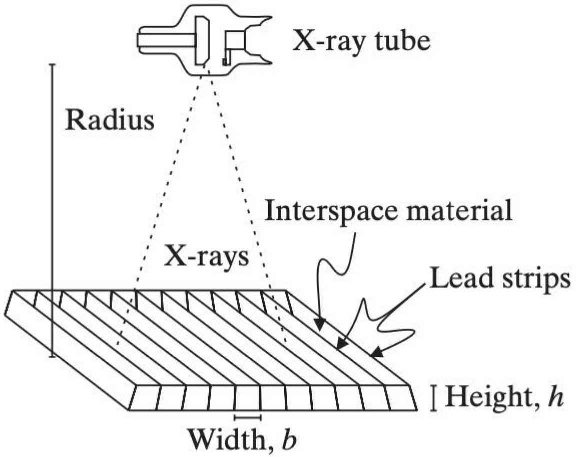
- Define the term *signal* and provide biologically relevant examples
- Demonstrate the importance of sampling signals above the *Nyquist Criterion*
- Implement signal processing techniques on time-series and image data

Signal Processing is Essential for Biomedical Imaging

- X-Ray Computed Tomography (CT) produces images of slices within the body

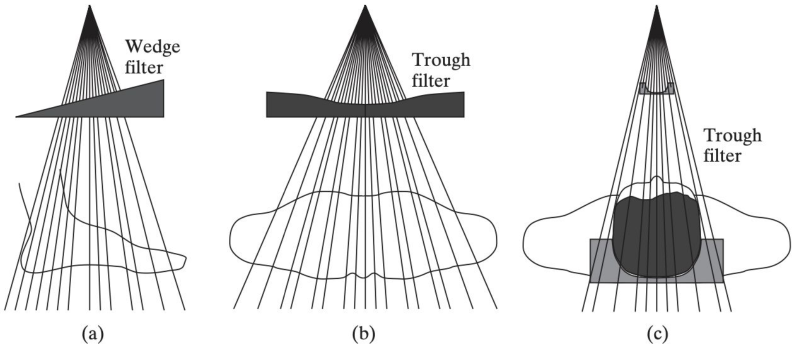
X-Ray Tubes Produce High Energy X-Rays

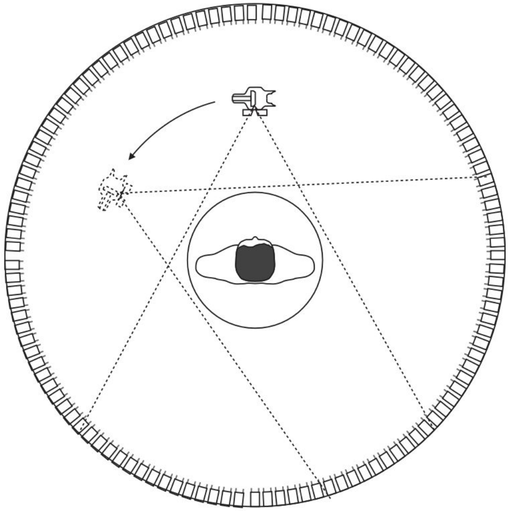
- X-Ray Computed Tomography (CT) produces images of slices within the body
- X-rays produced in the X-ray tube travel to detectors on the opposite side of the scanner



X-Ray Attenuation in CT

- X-Ray Computed Tomography (CT) produces images of slices within the body
- X-rays produced in the X-ray tube travel to detectors on the opposite side of the scanner
- X-rays passing through the body are differentially attenuated tissues





Sensors capture signals traveling in the same direction

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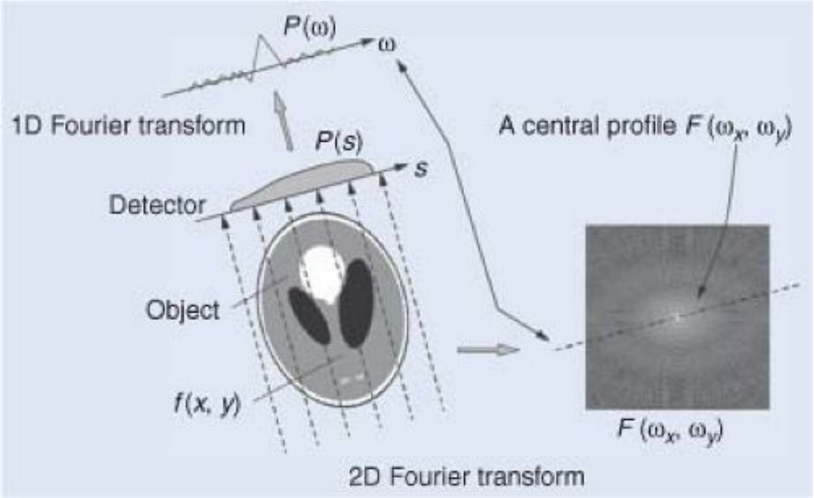
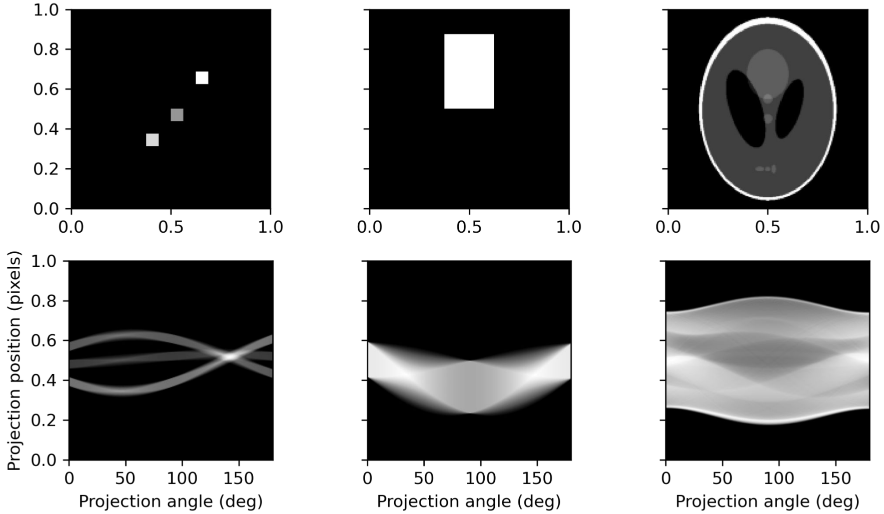
- X-rays produced in the X-ray tube travel to detectors on the opposite side of the scanner
- X-rays passing through the body are differentially attenuated tissues
- Attenuation along each radial direction is basis for CT signal



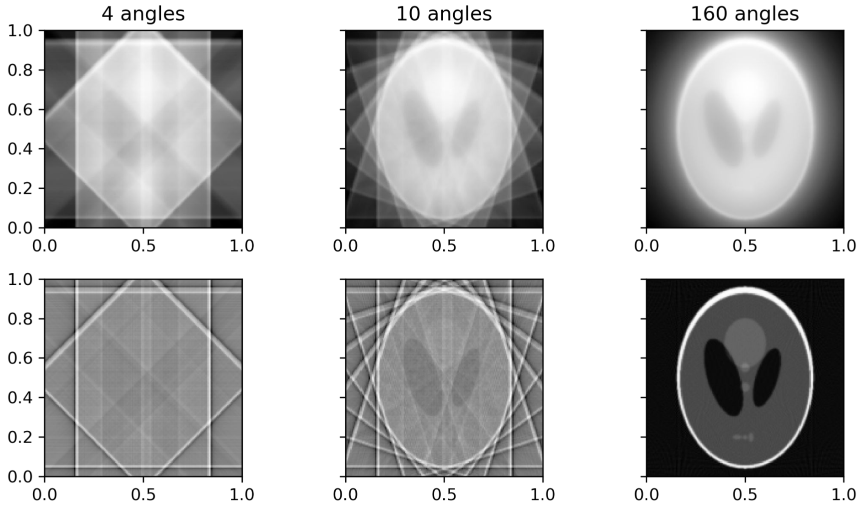
Sinograms summarize attenuation over all angles

- Attenuation values are measured for X-rays traveling over a range of angles
- Measurements at each sensor for each projection angle are captured in the Sinogram (Radon Transform)



Backprojecting Filtered Signals Recovers Scanned Object

- Attenuation values are measured for X-rays traveling over a range of angles
- Measurements at each sensor for each projection angle are captured in the Sinogram (Radon Transform)
- Scanned objects are recovered with Filtered Backprojection
- Having more measurement angles yields images with fewer artifacts



Signal Processing helps us analyze complex biological signals

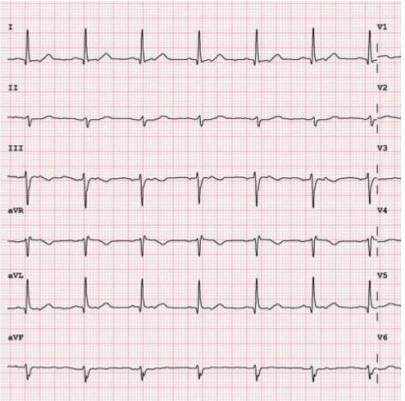
- Signal processing is the method of analyzing and modifying signals—such as sounds, images, or biological data—to make them clearer, more useful, or easier to interpret
- It involves techniques to filter out noise, enhance important features, or extract meaningful information from raw data
- In biology, signal processing can be used to clean up ECG recordings, improve the quality of MRI images, analyze DNA sequences, and much more!

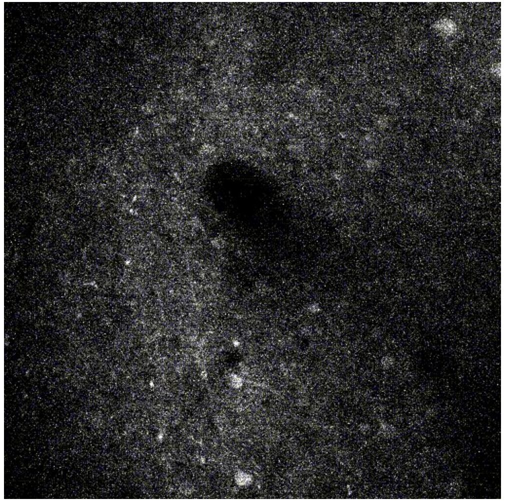
Part I: Signals & Sampling

What is a signal?

- A signal is any data or information that can be measured and represented in a way that changes over time or space

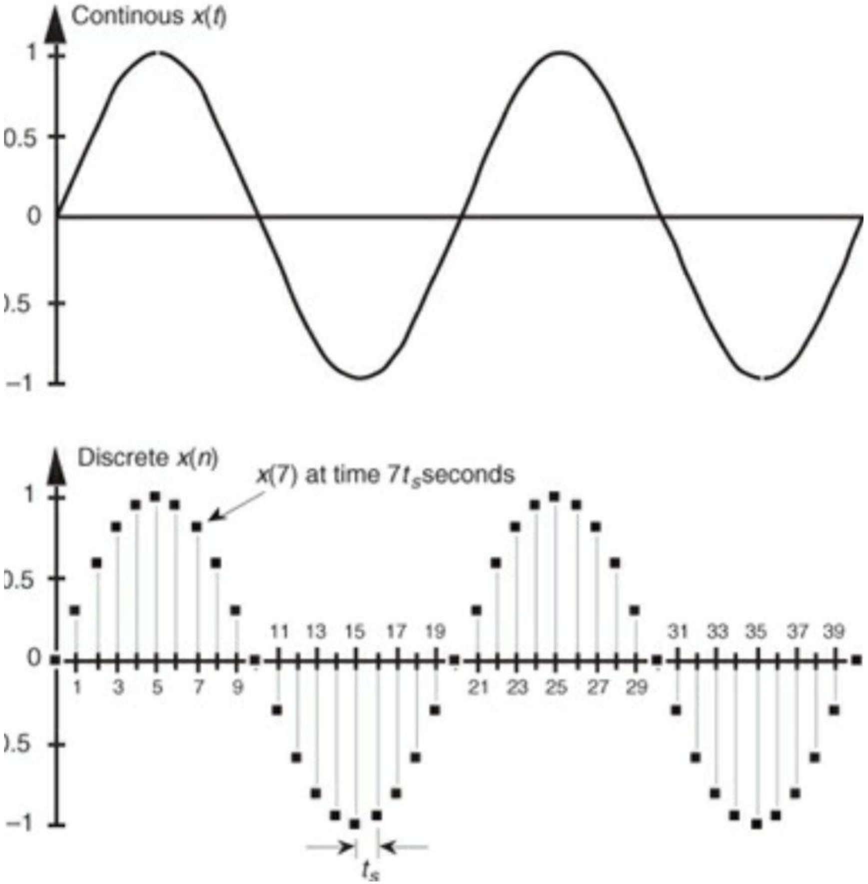
- Real-world signals have structure which contains meaningful information





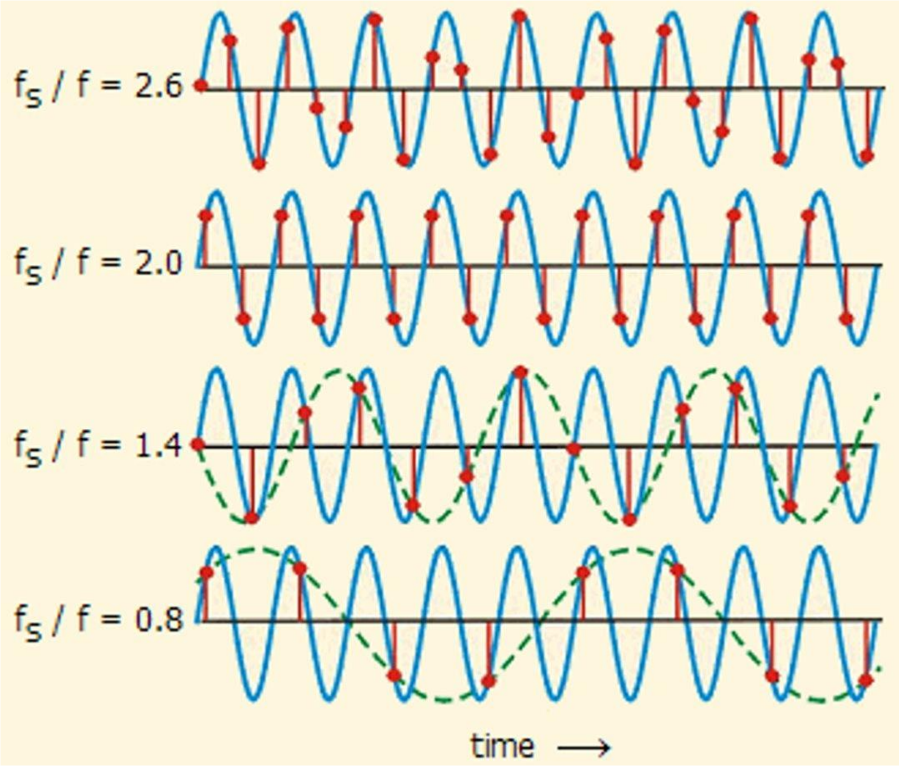
What is Sampling?

- A **time-series** is a signal that varies in time
- Many real-world signals are continuous time-series (e.g. heart beat, membrane voltage)
- To analyze continuous signals with a computer, we have to *discretize* them by sampling at specific time points



Nyquist Criterion and Sampling Rate

- We throw away information by sampling, so choosing an appropriate sampling rate is important
- The **Nyquist Sampling Criterion** tells us that we must sample our signal *at least twice as fast* *as the highest frequency in our data*



Let’s try an example in Datahub!

Part II: Time Series Signal Processing

Anything recorded continuously over time is a **time series**

(a set of data points generated from successive measurements over time)

Commonly encountered time series data in biology

- Gene expression data over time
- Neurophysiology recordings (e.g. electrophysiology, imaging)
- Circadian rhythm data
- Medical observations over time
- Animal movement
- Physiology data (e.g. heart rate/ECG, pulse rate, respiration, etc.)
- Molecules/proteins/cells moving

White blood cell tracking bacteria
Image info

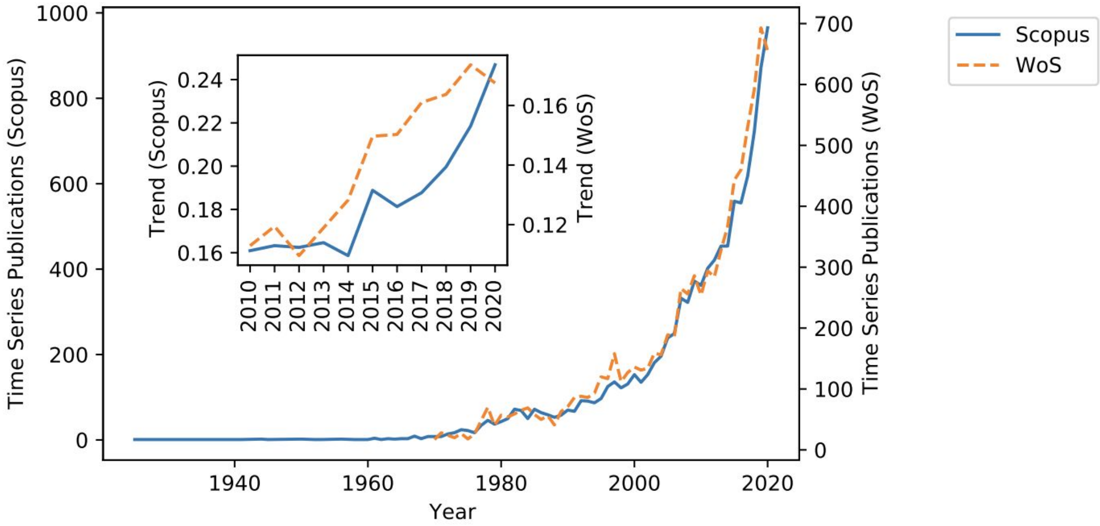


Fig. 1: Number of documents retrieved by Scopus (left axis) and Web Of Science (WoS, right axis) with the search string (restricted to documents titles): "time series" AND ("analysis" OR "data mining" OR "machine learning") over time. The inlet represents the trend (in %) over the last 20 years (the trend is normalized over the total publications in DBLP (the data is accessible in `https://dblp.org/statistics/publicationsperyear.html`).

More and more people developing time series analyses! (Siebert et al., 2021)

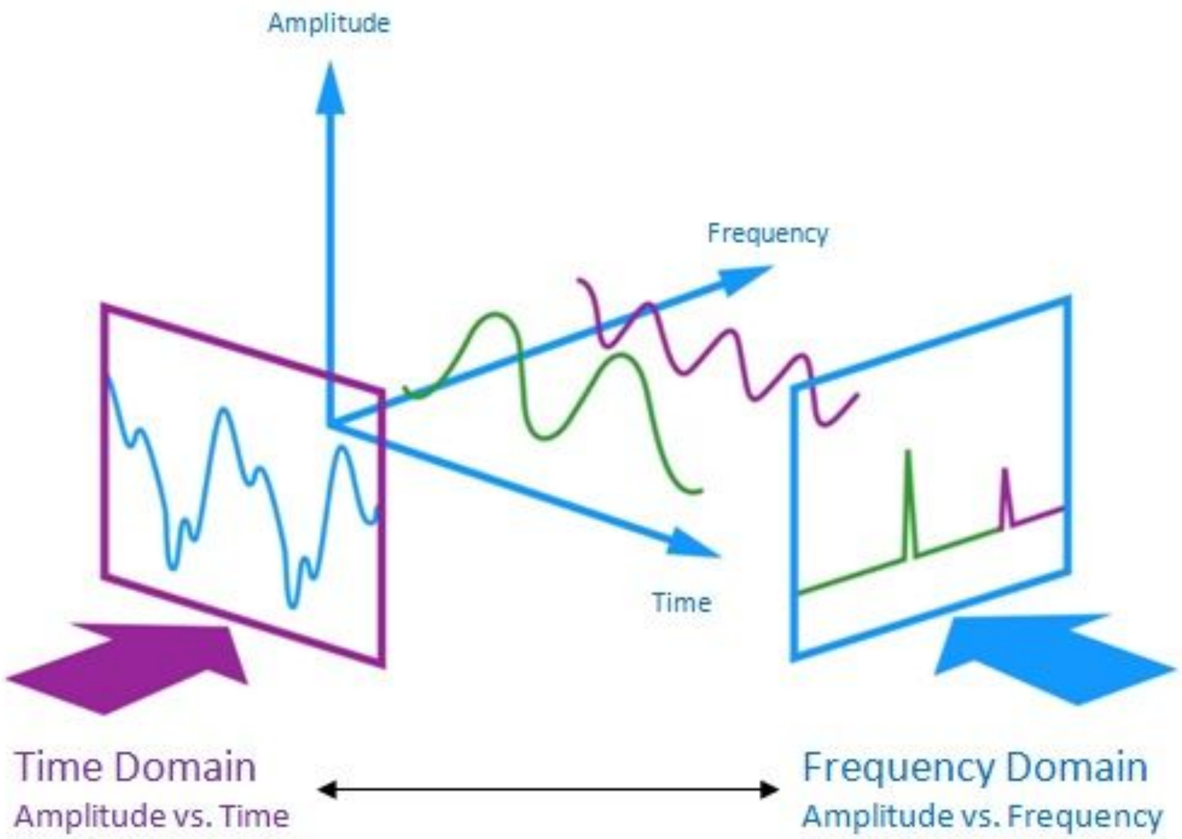
Common signal processing approaches

- Preprocessing & data cleaning
  - Removing outliers and/or noise
- **Filtering**
  - **High, low, bandpass**
- Looking for correlations in time
- Clustering & classification
- Dimensionality reduction or segmentation
- Prediction
- Anomaly or peak detection

Let’s give some a try in Datahub!

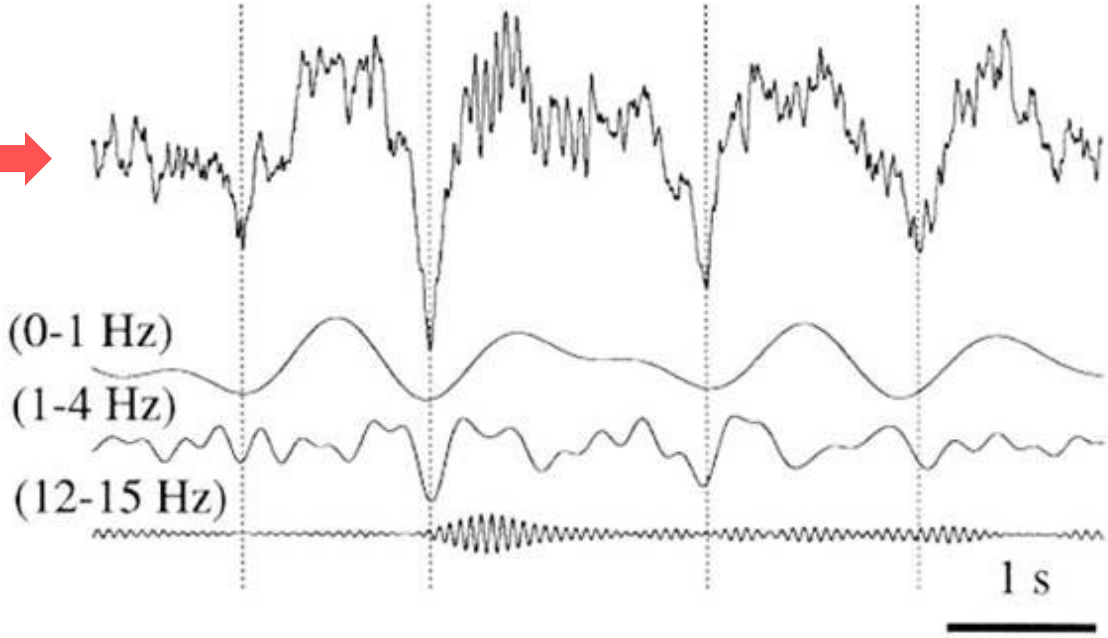
If a signal oscillates, we’re often interested in the **frequency** of these oscillations, rather than the signal itself.

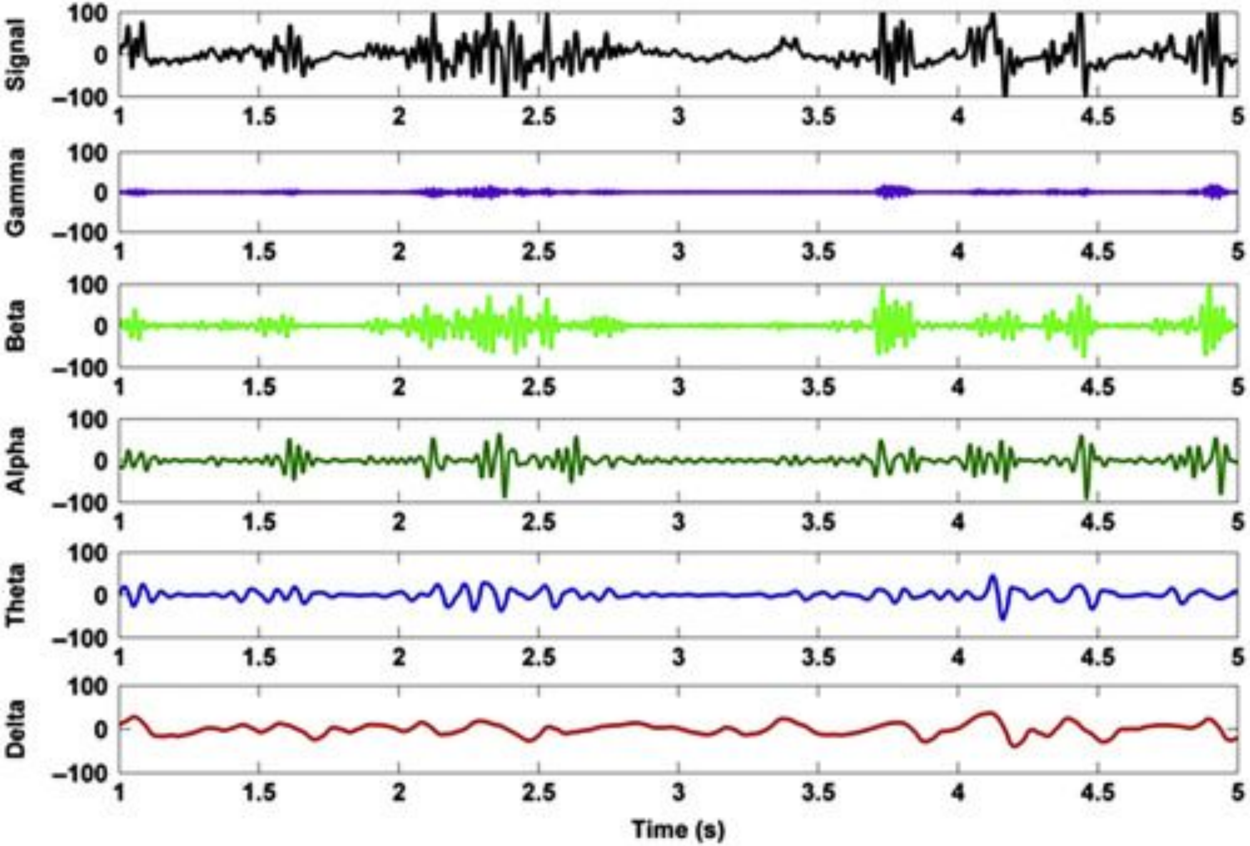
The goal: transform our data from the **time domain** to the **frequency domain**



Fourier transforms can decompose EEG signals & determine the power at specific frequency bands

original EEG signal





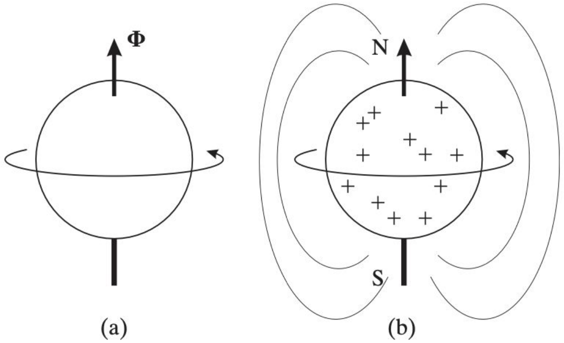
EEG Waves can be broken down into component frequencies

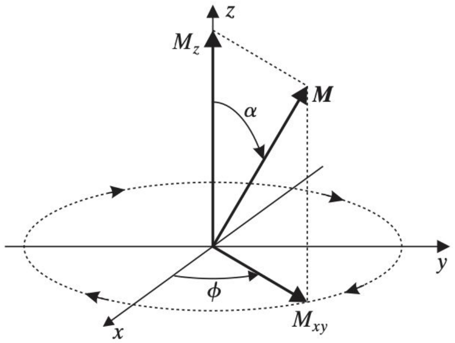
Magnetic Resonance Imaging (MRI) relies on Fourier Transforms of magnetic signals from stimulated tissues

- Produces high quality images without ionizing radiation
- Anatomical MR: T1-, T2-, Proton Density (PD)-, Diffusion-weighted
- Physiological MR: Arterial Spin Labeling (ASL), fMRI
- MRI systems include coils for applying magnetic & radio frequencies (RF) fields and coils that receive raw signal data

Applied Magnetic Fields to Nuclear Spin Systems Generates MR Signal

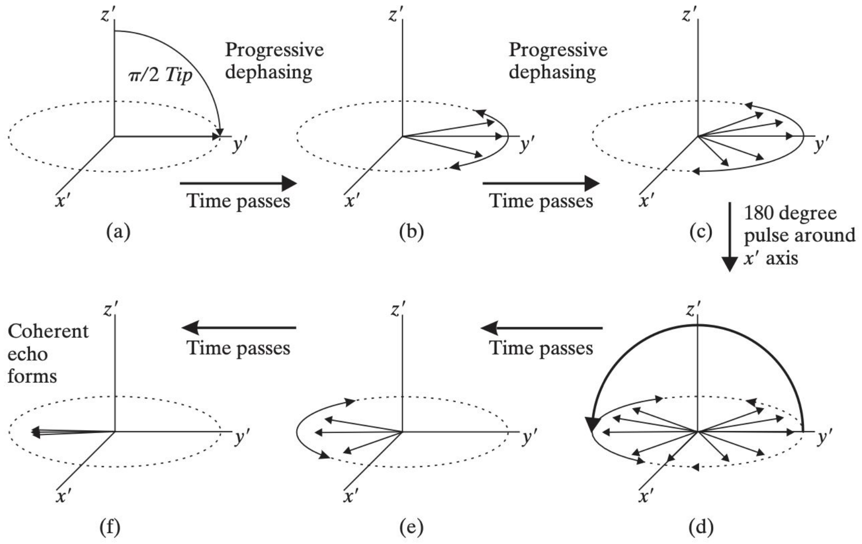
- Hydrogen atoms in our body form a **nuclear spin system**
- Each atom possesses a magnetic field direction in a random direction
- Applying an external magnetic field temporarily aligns the systems magnetic fields
- Processing magnetization vector produces basis for raw signal





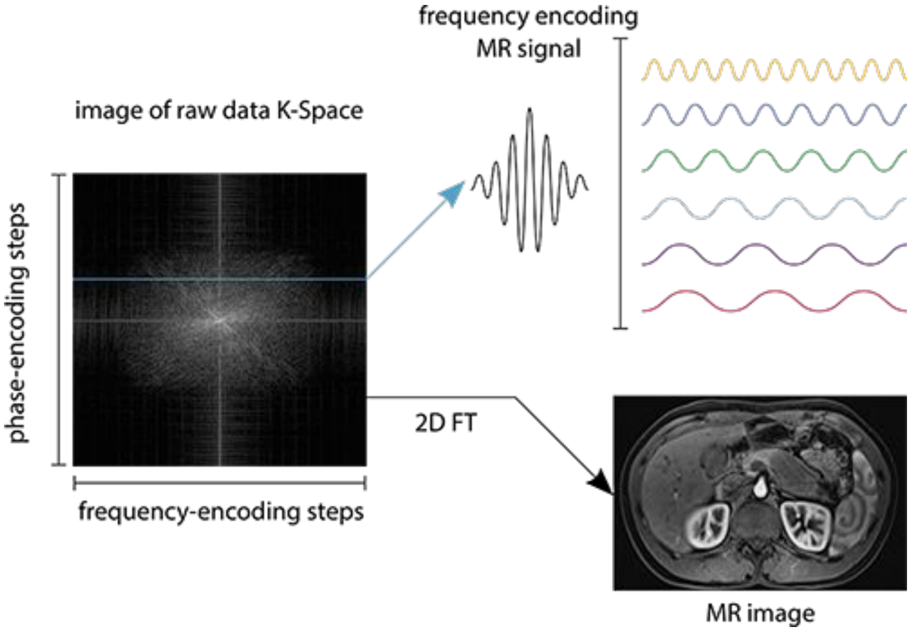
Manipulation of RF Pulses Creates Contrast in Different Tissues

- Following magnetization, hydrogen atoms dephase in a process called relaxation
- Relaxation time of transverse (T2) and longitudinal (T1) signals differs across tissues
- Various RF pulse sequences exist leverage differences to visualize specific tissues while masking others



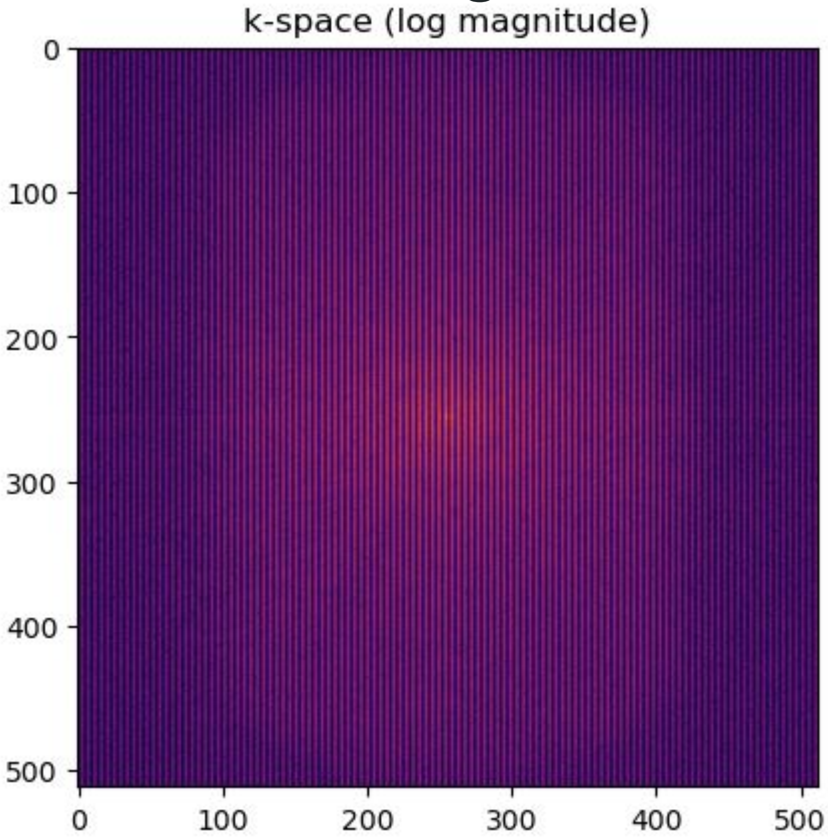
Fourier Transform Converts Raw Signal Frequencies stored in K-Space into an Image

- Sampled signal is stored in K-space:
  - Each point is a spatial frequency
  - Each pulse sequence adds a new row of phase-encoding space
  - Center: Low Frequencies, Contrast
  - Outer: High Frequencies, Details, Noise
- 2D Fourier Transform is applied to K-space signals to recover MR image



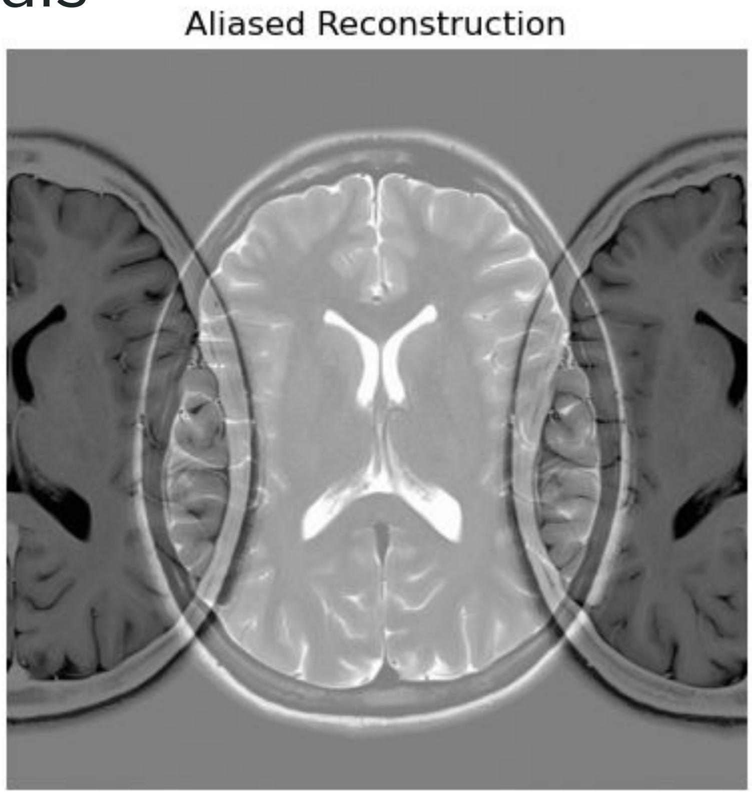
Undersampling in K-space Produces Aliasing

- Just like in the 1D case, we have to worry about undersampling
- We can simulate undersampling by setting every other column in k-space to 0



MRI: Undersampling in 2D signals

- Just like in the 1D case, we have to worry about undersampling
- We can simulate undersampling by setting every other column in k-space to 0
- Once transformed, we that high frequency signals wrap around the left/right edges while low frequency signals remain centered

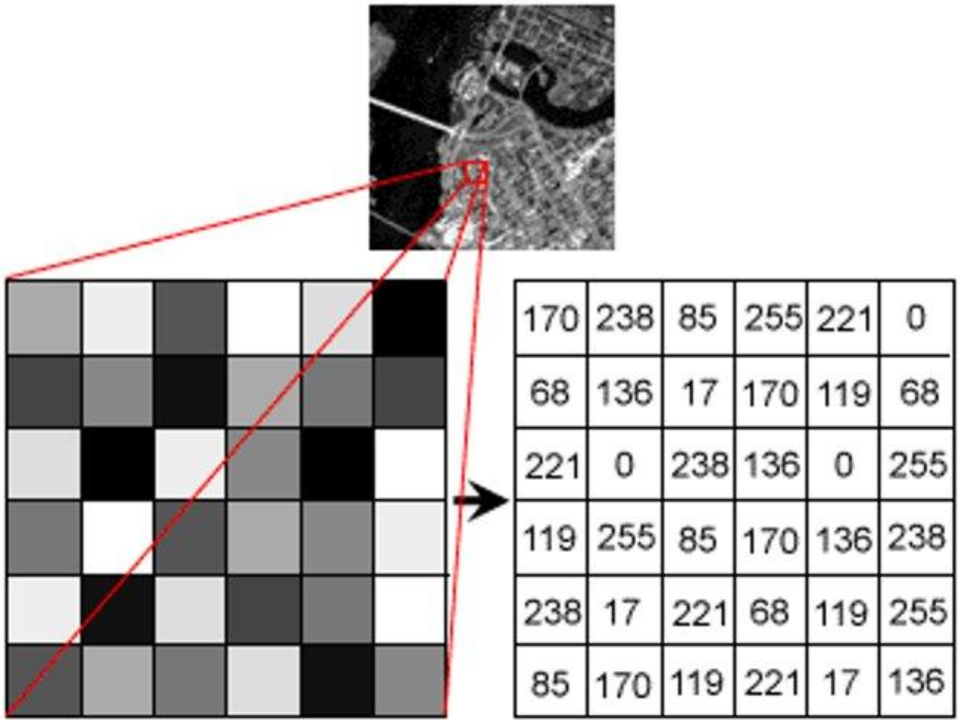


Let’s plot some examples in Datahub

Part III: Image Processing

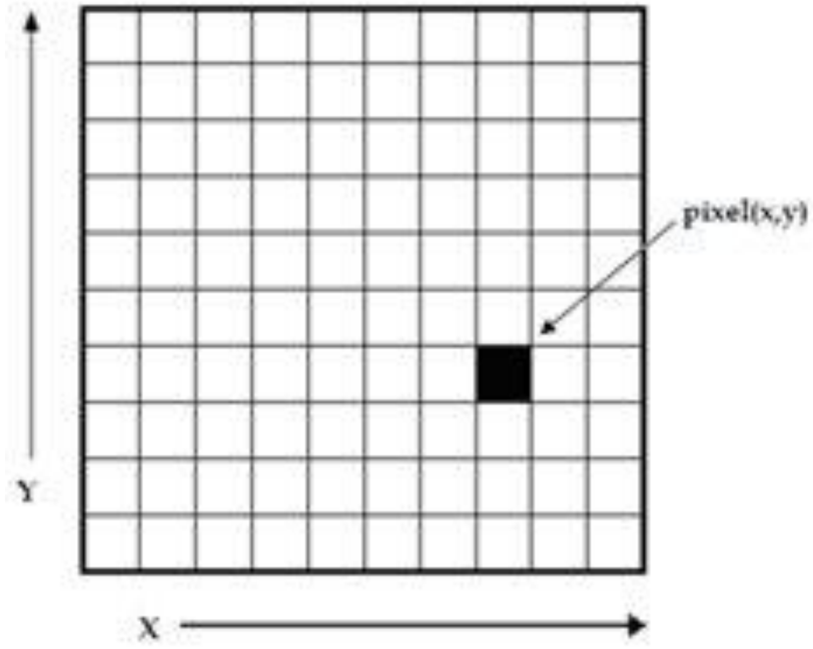
What is an image?

- An image is a collection of signals that vary spatially arranged on a 2D pixel grid



What is an image?

- An image is a collection of signals that vary spatially arranged on a 2D pixel grid
- Pixels are indexed with an ordered pair (x,y) and values increase **left→right** and **top→bottom** (just like a matrix)
- By convention the top, left corner is indexed as (0,0)
- Signals are contained within pixels and can be a single value (gray-scale)...



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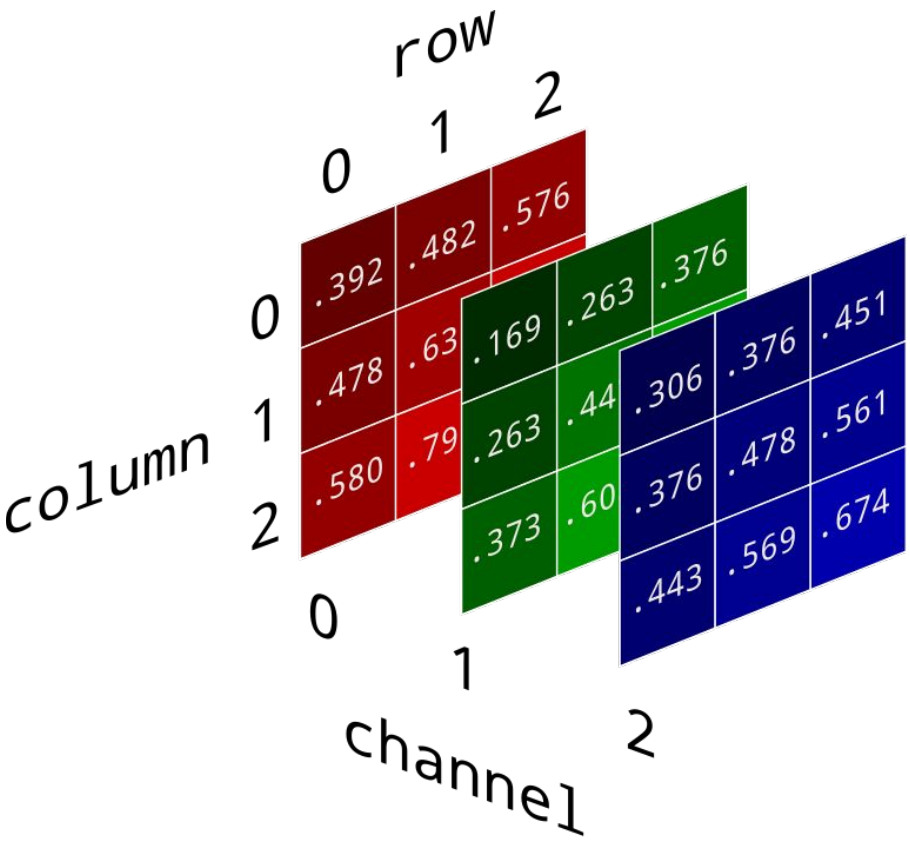
  Or a vector (RGB)!



Spatial Frequency Captures Repeated Image Structure

- Describes how patterns repeat per unit distance in an image
- Usually measured in cycles/ cm or cycles/ mm
  - Low-spatial frequencies change slowly, capture coarse features
  - High-spatial frequencies change quickly, capture fine details **and noise**

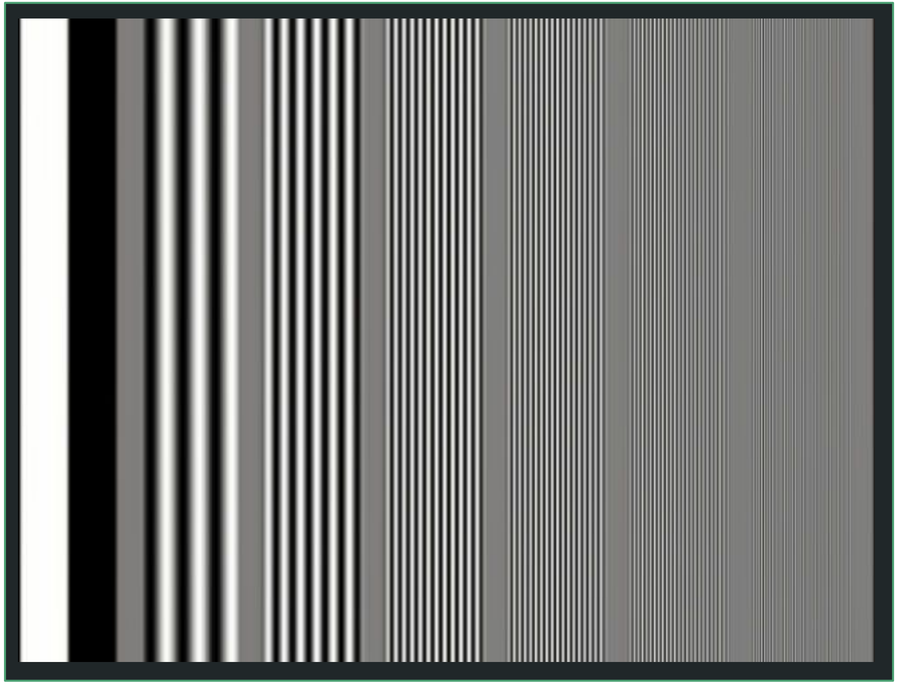
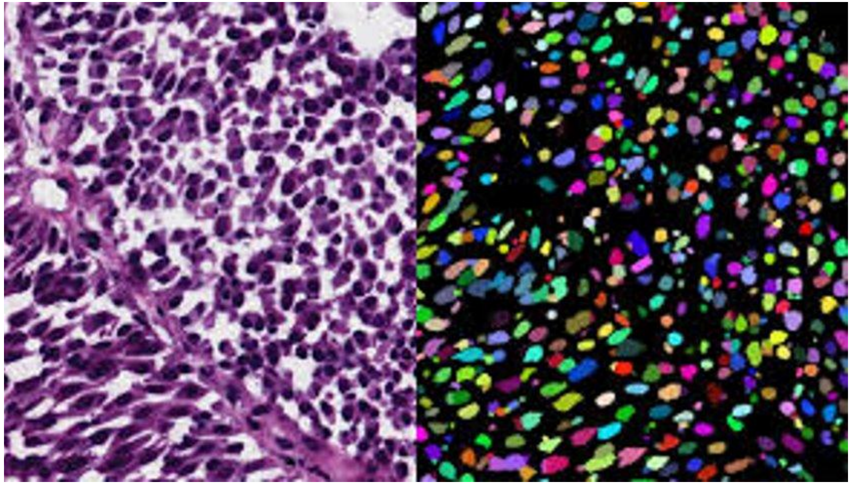
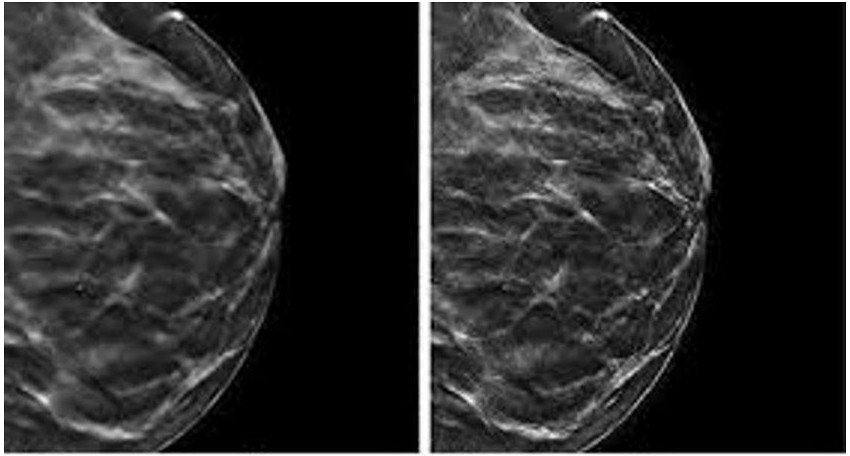


Image Processing

- Image processing can be used to extract or enhance useful features from image data
  - Isolate cells from background with image segmentation
  - Image enhancement with filters
  - Edge detection





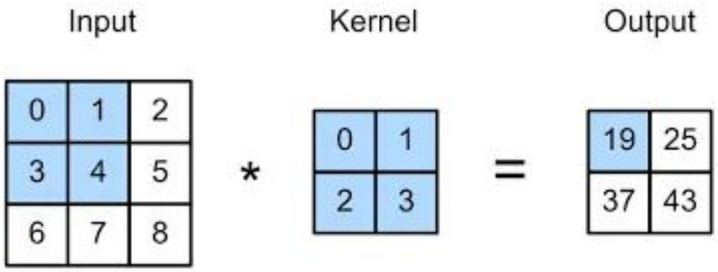
Before Sharpening After Sharpening

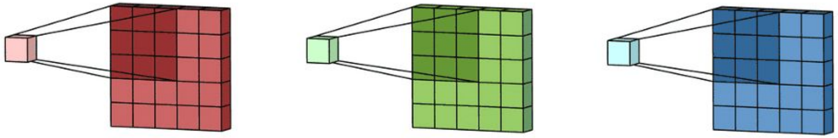
Image Processing

- Image processing can be used to extract or enhance useful features from image data
  - Isolate cells from background with image segmentation
  - Image enhancement with filters
  - Edge detection
- Let’s try applying some filters to this image from the scikit-image library using **convolution**

Convolutions

- Convolution is a common technique in image processing technique used to detect patterns, blur images, and detect edges
- It forms a basic building block for more advanced image processing and computer vision (e.g. CNNs)

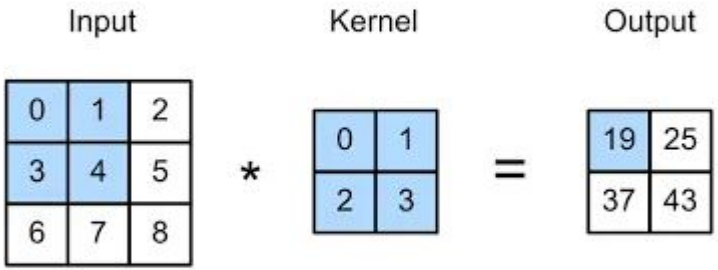


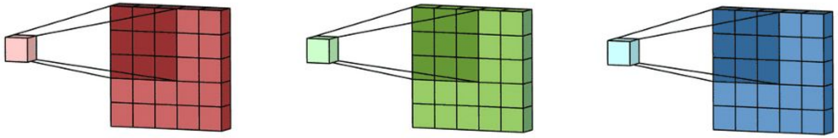


Convolutions

Convolution in 3 Steps

1. A small matrix (kernel) applied to part of the image
2. Kernel and pixel values are multiplied and summed to get a new pixel value
3. The kernel slides across the image, and the process repeats





Smoothing with Gaussian Kernels

- Gaussian Kernels act as **low-pass** filters (they let the low frequency signals pass through!)
  - Removes high-frequency – often used for noise reduction and blurring
- Convolution results in an average over image pixels with *a bias toward the pixel in the center* of the kernel
- Relative insensitivity to edges allows for smooth blurring which increases with kernel size
- **Reduces the spatial resolution** of convolved regions

$$\frac{1}{16}\begin{array}{|c|c|c|}\hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline\end{array}$$

Edge Detection

- Convolution with Sobel kernel detects edges in images
- Formed by outer product between two 1-D kernels:
  - Discrete derivative: [-1, 0, 1]
  - Binomial smoothing: [1, 2, 1]
- G_x and G_y find edges in x and y directions respectively
  - Add convolved images to recover all image edges

-1	0	+1
-2	0	+2
-1	0	+1

Gx

+1	+2	+1
0	0	0
-1	-2	-1

Gy

Let’s try this out in Datahub!

Additional Resources

Signal Processing

- https://mark-kramer.github.io/Case-Studies-Python/03.html
- https://voyteklab.com/oscillations/publications/interpreting-spectrum/

Image Processing

- 95 - What is digital image filtering and image convolution?
- Finding the Edges (Sobel Operator) - Computerphile
- Computer Vision Tutorial: A Step-by-Step Introduction to Image Segmentation Techniques (Part 1)

Classes at UCSD

- COGS 118C. Neural Signal Processing
- DSC 120. Signal Processing for Data Analysis